



OSTBAYERISCHE
TECHNISCHE HOCHSCHULE
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AI-based Anomaly Detection for Electric Vehicle Supply Equipment using LSTM Architecture

An der Fakultät für Informatik und Mathematik der
Ostbayerischen Technischen Hochschule Regensburg
im Studiengang
Wirtschaftsinformatik

eingereichte

Bachelorarbeit

zur Erlangung des akademischen Grades des
Bachelor of Science (B.Sc.)

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Abgabedatum: 23.08.2025

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Regensburg, den July 23, 2025

Yannic Hanel

Abstract

AI is important ...

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I List of Abbreviations

Electric Vehicle Supply Equipment (EVSE)
Electric Vehicles (EVs)
Charging Stations (CS)
Artificial Intelligence (AI)
Isolation Forest (iForest)
Recurrent Neural Network (RNN)
Exploratory Data Analysis (EDA)
Deep Neural Network (DNN)
Application Programming Interface(s)(APIs)
Information Technology (IT)

1 Introduction

1.1 Motivation and Relevance of Anomaly Detection in Electric Vehicle Supply Equipment (EVSE)

Electric Vehicle Supply Equipment (EVSE) has gained significant importance in recent years [EVSE24]. Analyzing the market sales of electric vehicles (EVs), we observe a remarkable growth: in 2012, only 120,000 EVs were sold, whereas by 2021, sales had surged to nearly 6.6 million. And to achieve net-zero CO₂ emissions by 2050, as planned by the European Union [EU2050], the number of EVs must continue to rise substantially—requiring an 11-fold increase from current levels by 2030 alone [IEA22].

To manage the much higher demand for EVs on our roads, we can assume that the market for EVSE will grow similarly to the resale of EVs. The correlation between the number of EVs and the number of EVSE is high. If we analyze the resale from the last few years, we can see that the top five countries in the European Union like Germany, Netherlands, France, and Belgium with the best EVSE infrastructure are also the top five with the largest market share of EVs [ACEA23].

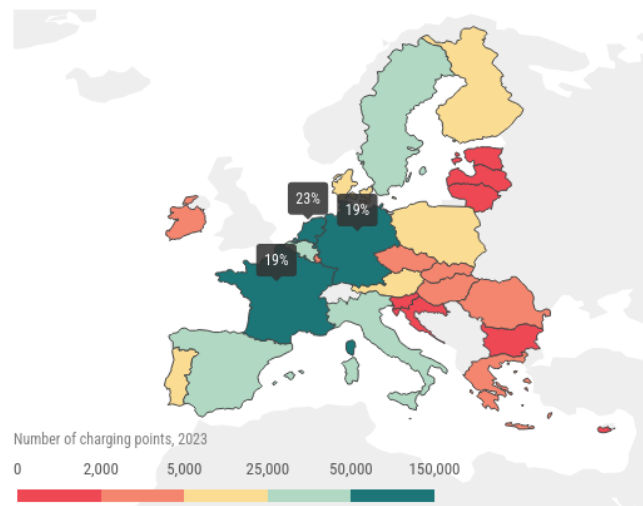


Figure 1: Availability of public charging points and market share of electric cars in the EU[ACEA23].

With the growing number of EVSE, particularly Charging Stations (CS), the interest of cyber-criminal groups in attacking these systems is also increasing. At first glance, EVSE systems may not appear to be direct targets for accessing critical data or manipulation. However, they can serve as entry points to gain access to more sensitive or private information [EntryPoints25]. The following figure illustrates how EVSE components, such as charging stations, can act as access nodes to critical or private data.

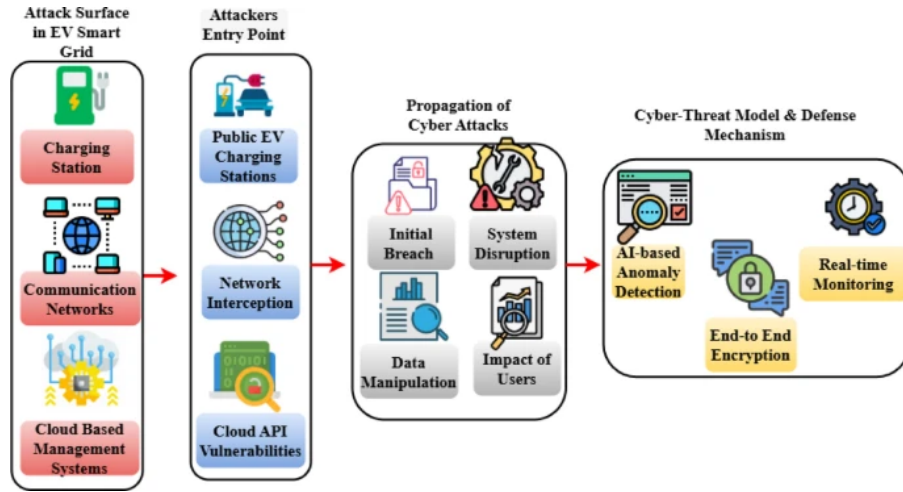


Figure 2: EVSE as an access node to critical data[EntryPoints25].

With the access to these kind of Data the critical attacks like data manipulations, data theft, distrub of service up to complete system failure are possible[EntryPoints25]. That does not only comes with a high financial risk, it also affect the trust of the users in the EVSE systems and potentially slowing down the adoption of EVs in general.

If we provide end users with a stable and secure EVSE infrastructure, consumer confidence in switching to electric vehicles will be significantly higher. This, in turn, ensures a consistent growth in the adoption of both EVs and EVSE systems[EVSE-Cyber25].

As seen in this first subchapter, the rapid growth of EVs and EVSE infrastructure also requires a robust monitoring and anomaly detection system to ensure the security of the EVSE infrasructure. Only if we can ensure the security of the EVSE infrastructure can we maintain user trust in the energy transition and avoid major cybersecurity headlines in the future.

Ensuring high security standards for EVSE systems is of critical importance, yet difficult to achieve. The EVSE infrastructure constitutes a complex system that includes not only charging stations but also sophisticated backend systems and communication protocols. These components and software solutions are typically developed, published, and maintained by various vendors, which makes securing the entire system a challenging task.

Combining all the different software from vendors into a secure infrastructure is challenging enough, but the EVSE infrastructure, especially charging stations (CS), is also regularly installed in public places. Even when not in a public setting, these installations are often outdoors, where security measures are significantly lower than in buildings protected by walls, doors, and other physical barriers.

In this paper, we focus on wallboxes, which are generally smaller than CS and are typically used in private settings. They are often installed directly on house walls or in garages, and thus still form part of the overall EVSE infrastructure.

The fact that they are installed in private areas initially suggests a higher level of security.

However, since these areas are still outdoors, we must assume that there are no additional physical security measures beyond the wallbox itself. Therefore, it is crucial to develop a system that not only protects against software-based attacks, such as Denial-of-Service (DoS) or Distributed Denial-of-Service (DDoS), but also considers other potential attack vectors such as firmware manipulation and unauthorized data interception. In addition, the system must secure the hardware against physical threats like tampering, theft, or other intrusions. In our project, we use wallboxes from the [ReSiLENT] project, provided by [StoehrMobility]. The software is developed by [ChargeIQ] and already includes basic security measures to protect against external threats. This project is also carried out in cooperation with [ASVIN]. In the following section, we will take a closer look at the internal components of the wallbox.

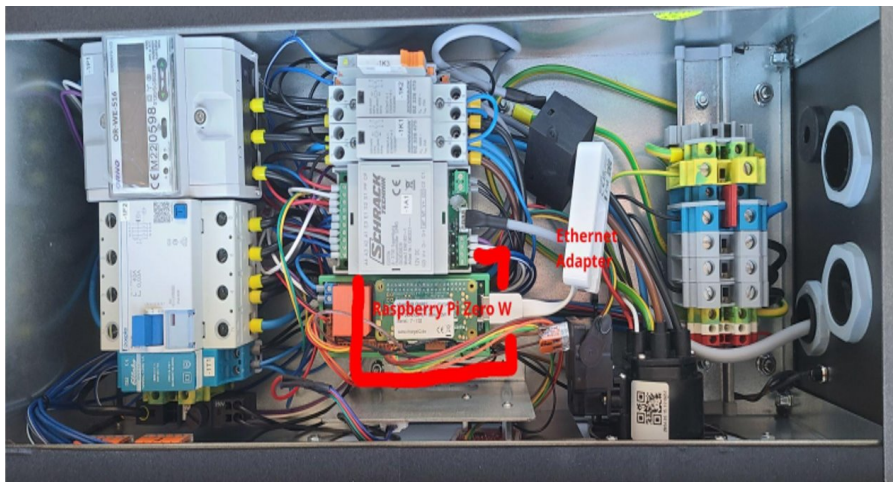


Figure 3: Interior view of the wallbox[Han25].

In this figure, the internal components of the wallbox are visible. The wallbox is equipped with a Raspberry Pi Zero W, which serves as the main controller for the planned peer-to-peer communication between the wallboxes. This device is also intended to host of the detection system developed as part of this project.

Located behind the Raspberry Pi Zero W is another Raspberry Pi, which is responsible for communication with the backend, including the transmission of user data and handling of payment processes from the Charging process. This component is secured and programmed by *ChargeIQ GmbH*[ChargeIQ].

The sensitive hardware components are enclosed in an aluminum casing designed by *Stöhr Mobility GmbH*[StoehrMobility], which provides a degree of protection against physical attacks. However, upon closer inspection of these physically secured components, it must be acknowledged that the protective measures can be relatively easily bypassed. This has already been observed and documented by our colleague Lea Laux in her work [Laux24].

As already mentioned, the wallbox also processes personal data of users, such as their user ID and payment information, in communication with the backend server. This requires special

attention in the context of the DSGVO / GDPR, as additional requirements for the protection of personal data like data minimization, storage limitation [GDPR5] and legal obligation [GDPR6] must be taken into account throughout the entire EVSE infrastructure.

Security requires significant computational resources, particularly in the domain of anomaly detection. As shown in Figure 3, the wallbox is equipped with a Raspberry Pi Zero W, which offers relatively sufficient resources for a single-board computer. However, when processing and analyzing over 1,000 hardware events to detect potential anomalies, the system quickly reaches the limits of its hardware capabilities this is just a small example of the challenges we face in securing EVSE infrastructure.

Due to the combination of digital and physical threats, the involvement of multiple vendors, and strict data protection requirements, securing Electric Vehicle Supply Equipment (EVSE) infrastructure is a complex task and brings a lot of Challenges.

1.2 Overview of AI-based approaches

Artificial intelligence is often used in different contexts, therefore we have to define what we mean by AI-based approaches and what different approaches exist. At first it is a leading technology in the current digital transformation and is crucial to move forward the fourth industrial revolution[AI-Definition].

The Goal of AI is to create machines that can perform tasks that typically require human intelligence, such as understanding natural language and recognize patterns in data. Especially detection of patterns is in this work from Relevance. We can divide the different AI-based approaches into two sub categories:

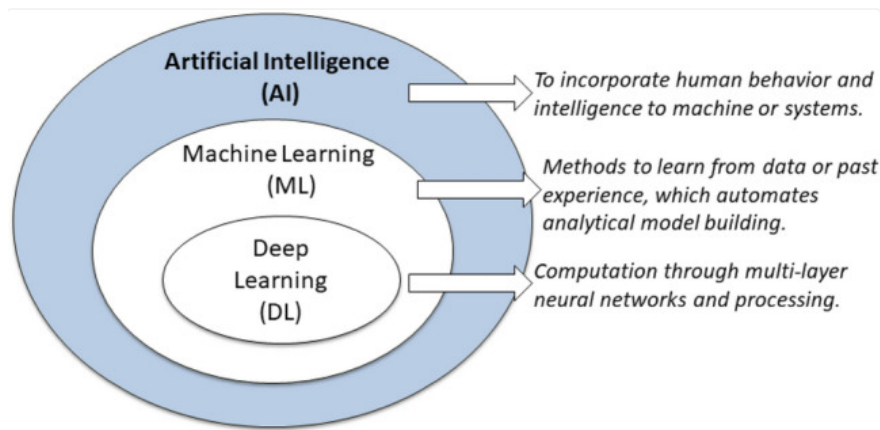


Figure 4: An illustration of the position of machine learning (ML) and deep Learning (DL) within the area of artificial intelligence (AI)[AI-Definition]

these can be further divided into three categories:

- **Supervised Learning:** This approach involves learning from labeled data to identify patterns and make predictions. It is commonly used in tasks such as email spam classification or price prediction, where the algorithm is trained on known input-output pairs.

- **Unsupervised Learning:** This data-driven method is used to uncover hidden patterns in unlabeled data. Techniques such as clustering and dimensionality reduction are applied to identify structures within the data, for example, to segment customer groups or discover business models.
- **Semi-Supervised Learning:** This approach combines aspects of both supervised and unsupervised learning by utilizing a small amount of labeled data alongside a larger pool of unlabeled data. It is particularly useful when labeling data is expensive or time-consuming, but large amounts of unlabeled data are available.

[AI-Definition]

In the following table, we categorize the different approaches alongside their corresponding machine learning architectures. This allows us to identify the advantages and disadvantages of each method and to gain a clearer understanding of which approach is most suitable for producing actionable results in real-world deployments.

Approach	Examples	Advantages	Disadvantages
Supervised Learning	Decision Trees, Random Forest, LSTM	High prediction accuracy, Clear objectives	Requires a lot of labeled data, Risk of overfitting
Unsupervised Learning	K-Means, Isolation Forest	Finds patterns without labels, Useful for unlabeled data	Hard to interpret, No clear target
Semi-Supervised Learning	Self-training	Improves performance with few labeled data	Complex to implement

Table 1: Comparison of machine learning approaches with examples, advantages, and disadvantages. Own illustration based on [AI-Definition].

After gaining an overview of the different machine learning approaches, we can now delve deeper into the field by examining various methods for anomaly detection and identifying the most effective models for detecting typical anomalies.

1.3 Research Questions

The following research questions guide this thesis on **AI-based Anomaly Detection for Electric Vehicle Supply Equipment using LSTM Architecture**:

- What types and characteristics of anomalies occur in electric vehicle supply equipment (EVSE)?

- How can relevant features be identified and selected to effectively train machine and deep learning models for anomaly detection?
- How does the performance of LSTM-based models compare to that of traditional or alternative AI-based methods in terms of detection accuracy and false positives?
- What practical challenges arise when implementing such anomaly detection systems in real-world charging infrastructure?

These research questions address both the technical and real-world aspects of developing and applying anomaly detection, providing the analytical framework for the subsequent chapters of this thesis.

1.4 Structure of the Thesis

This thesis is divided into ten chapters, each building upon the previous to explore the use of AI-based methods—specifically LSTM architectures—for anomaly detection in Electric Vehicle Supply Equipment (EVSE).

Chapter 1 introduces the motivation and relevance of the topic, outlines the role of anomaly detection in the context of EV infrastructure, and formulates the central research questions. Additionally, it provides an overview of prior AI-based approaches and concludes with this structural outline. Chapter 2 summarizes the key theoretical foundations necessary for the thesis, including the basics of anomaly detection, a short introduction to tree-based models such as decision trees and random forests, and a focused explanation of the LSTM architecture.

Chapter 3 presents related work, covering research on security aspects and anomaly detection mechanisms specific to EVSE, as well as a broader review of AI-based detection methods applied in similar domains. Chapter 4 describes the dataset creation and acquisition process, detailing both public datasets and custom setups involving Prometheus monitoring tools. Chapter 5 follows with an exploratory data analysis (EDA), intended to uncover key patterns and differences between the public and self-generated datasets.

In Chapter 6, the process of data preprocessing and feature selection is discussed, including the identification of relevant features and the use of feature importance methods such as decision trees. Chapter 7 focuses on the LSTM model itself, describing its architecture, the training process, and the chosen strategies for hyperparameter tuning.

The results of the approach are then evaluated in Chapter 8, which outlines the experimental setup, evaluation metrics, and key results of the applied anomaly detection system. Chapter 9 provides a detailed discussion of the findings in relation to the research questions, highlighting limitations and potential practical applications. Finally, Chapter 10 concludes the thesis with a summary of the work and a reflection on future directions.

This structure ensures a logical and methodical development of the topic, from theoretical foundations to implementation, evaluation, and critical interpretation.

2 Fundamentals and Key Terminologies

2.1 Basics of Anomaly Detection

In the previous section, we took a closer look at what Artificial Intelligence is and how useful it can be in the field of EVSEs. In this section, we delve deeper into the area of Anomaly Detection and its fundamental concepts.

Anomaly Detection is the process of analyzing data patterns to identify outliers — in other words, data points that deviate significantly from the norm. This technique can be applied in a wide range of domains, including credit card fraud detection, insurance and healthcare monitoring, intrusion detection in cyber-security, fault detection in safety-critical systems, military surveillance, and also in the field of EVSEs.[BAN09]

Outliers are data points that differ significantly from the majority of the data. In the paper by Hodge and Austin, they define outliers as “data points that are inconsistent with the rest of the data.” The paper also includes some informative graphics that illustrate the different types of outliers and how they can be detected.[HA04]

Type 1 is the most common form of outlier, where data points are generally clustered around

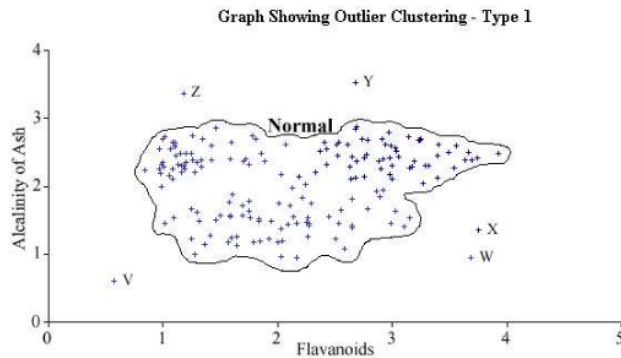


Figure 5: Data distribution classified by Type 1 – Outlier Clustering. The data is adapted from the Wine dataset (Blake and Merz, 1998)[HA04].

a central value, but a few points differ significantly from the rest. This can be seen in the figure above, where most data is centered, yet a few outliers deviate clearly.

Such outliers can be detected using a simple z-score algorithm, where the mean μ and the standard deviation σ of the data are calculated. The z-score is then computed using the formula: $Z = \frac{X - \mu}{\sigma}$. Estimated z-scores greater than 3 or less than -3 can be considered anomalies. These anomalies should be given increased attention during data analysis. Additionally, anomalies can be contextualized by parameters such as time or location to improve their identification.[Per19]

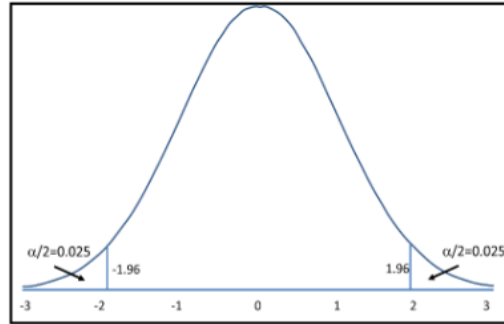


Figure 6: Z-score with critical region[Per19]

But before you can calculate the z-score you need an extensive data collection and preprocessing. This is also the first step in the Anomaly Detection process. The data must be cleaned, normalized, and transformed to ensure that it is suitable for analysis. This may involve removing duplicates, handling missing values, and scaling the data to a common range.

After the data is preprocessed, feature engineering is performed to extract relevant features from the data. Feature engineering involves selecting and transforming the data; this can sometimes also create new features that can improve the performance of the anomaly detection model. Also, data reduction is a relevant step in this process, where the data is reduced to a smaller set of features that are most relevant for the anomaly detection task.[Mah22]

If you are done with the data collection and preprocessing, you have to select a suitable model for your case of anomaly detection. But before we analyse the different models, let's take a look at the typical challenges of anomaly detection.

Like mentioned before to detect anomalies we have to analyze data patterns to identify outliers. But to determine the correct border between normal and abnormal data can be challenging and depends extremely on the context of the data.

The EVSE infrastructure is changing rapidly, not just with new technologies the charging stations change with every new charging process and communication with other CS in the network. This makes it difficult to define a static outlier threshold. Instead it's extremely important to adapt the model to the current state of the EVSE network and the charging process to generate dynamic detection in all situations in the dynamic EVSE environment.[BAN09]

Before the z-score can be calculated, extensive data collection and preprocessing are required. This forms the first step in the Anomaly Detection process. The data must be cleaned, normalized, and transformed to ensure it is suitable for analysis. This may involve removing duplicates, handling missing values, and scaling the data to a common range.

After preprocessing, feature engineering is performed to extract the most relevant characteristics from the dataset. Feature engineering involves selecting and transforming the data — and sometimes even creating new features — to improve the performance of the anomaly detection model. Additionally, data reduction is a crucial part of this process. It focuses on reducing the dataset to a smaller subset of features that are most relevant to the anomaly detection task.[Mah22] This step is of high relevance for the dataset used in this work, with further details

provided in the *Data Acquisition and Preprocessing* chapter.

Once data collection and preprocessing are completed, the next step is to select a suitable model for the specific anomaly detection use case. But before we analyze the different models, it is important to examine the typical challenges associated with anomaly detection.

As previously mentioned, detecting anomalies requires analyzing data patterns to identify outliers. But determining the line between normal and abnormal behavior is challenging and highly dependent on context.

The EVSE infrastructure is also evolving rapidly — not only due to new technologies, but also because charging stations behave differently during each new charging process and through communication with other charging stations in the network. This variability makes it difficult to define a static outlier threshold. Therefore, it is essential to adapt the anomaly detection model to the current state of the EVSE network and the ongoing charging processes. This enables dynamic detection capabilities that remain effective under all conditions in a highly dynamic EVSE environment.[BAN09]

2.2 Introduction to Decision Trees, Random Forest, Isolation Forest and LSTM

In this section, we will provide a brief introduction to several machine learning models who are categorized as supervised learning models. These models are also the most relevant for our project.

The **Decision tree** classifiers are the most well known and one of the powerful models in various fields. It works by creating a tree-like structure with Nodes and Branches and every Node is an Attribute or feature and the Branch can refer to the characteristic of the Attribute[DecisionTreeExplain24].

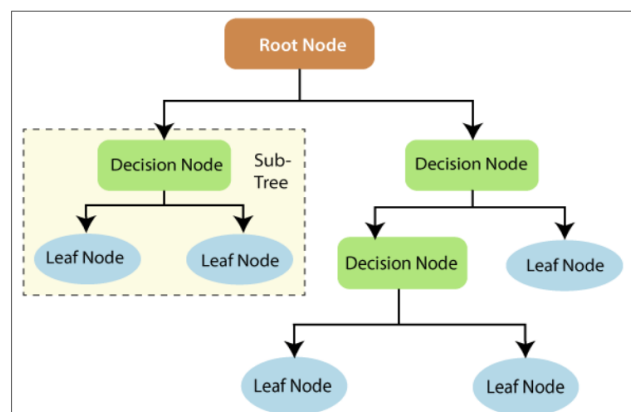


Figure 7: Decision Tree[DecisionTreeExplain24]

The decision rule is not a black box and can be easily interpreted. It compares the value of the feature with a threshold and then decides which branch to follow. It is not only used in

Machine Learning, but is also common in fields like image processing and pattern recognition, due to its simple analysis and high accuracy. Despite these advantages, it has a high risk of overfitting, especially with small datasets.[DecisionTreeExplain24]

The **Random Forest** is an ensemble method that combines multiple decision trees that work independently of each other. Each decision tree is created using a random vector to improve variance and reduce overfitting.

The final decision is made by majority vote of all the trees together. This method is more robust against overfitting and can handle large datasets with high dimensionality. Two key vectors influence the performance of the Random Forest: the correlation between the trees and the strength of the individual trees.[Bre01]

The **Isolation Forest** is an innovative approach for anomaly detection. While conventional methods typically create a model of "normal" behavior and classify deviations from it as anomalies, the iForest follows the principle of directly isolating anomalies.[LTZ08]

The explanation for this is that anomalies are characterized by their rarity and by being significantly different in their attributes. Therefore, they are easier to separate from the majority. This model brings advantages such as robustness with regard to dimensionality and irrelevant features. On the other hand, *swamping* is one of the disadvantages, where normal instances that are too close to anomalies can prevent anomalies from being isolated and thus make them harder to detect.[LTZ08]

A **Long Short-Term Memory (LSTM)** is a special type of Recurrent Neural Network (RNN) designed to model sequential data and long-term dependencies. Especially the long-term dependencies have high priority in this project, where the focus is on detecting anomalies over longer time periods. LSTMs use memory cells and gating mechanisms to store information over long time steps reliably. Despite their many advantages, they come with many different parameters, making them difficult to tune. Additionally, they represent a black-box model, meaning it is hard to understand why the model made a certain decision.[SM19]

Machine Learning Model	Advantages	Disadvantages
Decision Tree	Easy to understand and interpret	High possibility of overfitting
Random Forest	Less overfitting than a single decision tree; handles high-dimensional data	Not well-suited for small datasets
Isolation Forest	Robust to high-dimensional and irrelevant features	Swamping effect in presence of nearby normal instances
Long Short-Term Memory	Effective at modeling long-term dependencies	Black-box nature and complex parameter tuning

Table 2: Comparison of supervised learning models with advantages and disadvantages. Own illustration based on [DecisionTreeExplain24][Bre01][LTZ08][SM19].

2.3 Overview of LSTM (Long Short-Term Memory) Architecture

testets

In this chapter, I want to provide a detailed overview of the LSTM architecture, as it is the most frequently used and complex model in my work.

Long Short-Term Memory, or LSTM for short, was developed in 1997 by Sepp Hochreiter and Jürgen Schmidhuber as a solution to the vanishing gradient problem that occurs in traditional Recurrent Neural Networks (RNNs) [SM19]. It allows the network to learn dependencies over much longer time sequences, which is particularly important in my datasets, where attacks can be insidious and not immediately obvious.

This addresses one of the core limitations of traditional RNNs, which struggle to learn long-term dependencies due to the vanishing gradient problem. When training these networks, the error signals either explode—causing oscillating weights and unstable learning—or vanish, preventing the network from learning dependencies over extended time periods. Standard RNNs can typically bridge about 5 to 10 time steps, whereas LSTMs are capable of learning dependencies over 1,000 or more time steps [SM19], LSTMs process inputs sequentially, with a constant runtime of $\mathcal{O}(1)$ per time step.

But how does this work, and how is it possible to achieve this with constant time complexity? The LSTM architecture consists of six different internal states that are processed in a specific sequence to achieve this goal.

- **Forget Gate:** The forget gate calculates a dynamic value between 0 and 1 that scales the previous cell state. A value of 0 means "forget everything" and a value of 1 means "keep everything". This allows the network to selectively forget information that is no longer relevant.
- **Input Gate:** The Input gate decide how much of the new information should be added to the new cell state at the current time step. It is a kind of protective mechanism that tries to prevent irrelevant inputs from being added to the cell state.
- **Candidate Values:** The candidate values are the new information that get in form of squased values that get written into the memory cell.
- **Cell State Update:** The cell state is updated by simultaneously scaling down the existing state via the forget gate and adding in the gated candidate values. This allows the network to maintain a stable memory while also incorporating new information.
- **Output Gate:** The output gate determines which parts of the cell state should be realy becomes the output of the LSTM and protecting in this way the network from sharing unnecessary information too early.
- **Hidden State:** The hidden state is the final output of the LSTM for the current time step and what the outpute gate allows to be shared with the next time step.

[SM19]

Machine Learning Model	Advantages	Disadvantages
Long Short-Term Memory	Excellent long-term memory capabilities	Fixed network topology – cannot dynamically adjust memory capacity
	Selective information processing through gates	Complexity – more parameters than simple RNNs
	Stable learning without gradient explosion/vanishing	Modularization challenges – unclear how to optimally structure networks

Table 3: Strengths and Limitations of LSTM. Own illustration based on [SM19].

3 Related Work

3.1 Recent Advances in Cybersecurity and Anomaly Detection for EVSE

In the emerging field of EVSE, numerous studies focus either on electric vehicle supply equipment itself or on anomaly detection methods. However, the specific combination of anomaly detection applied to EVSE is not yet widely explored. Therefore, the paper [EDBF24], titled *Enhancing EV Charging Station Security Using a Multi-dimensional Dataset*, is highly relevant to this research. The authors present a dataset used for anomaly detection that includes similar features such as hardware events, power consumption, and other HCP-related data. The paper focuses on various attack scenarios and attempts to detect them using several machine learning models.

Another relevant contribution in this field is the paper [EVSE24], titled *Regression Based Anomaly Detection in Electric Vehicle State of Charge Fluctuations Through Analysis of EVCI Data* which not only highlights the importance of anomaly detection in EVSE, but also provides a comprehensive overview of a realistic charging station scenario. The setup includes four different charging boards, six charging ports, and a battery energy storage system (BESS) handling various types of EVs. The study monitors the state-of-charge (SoC) values of EVs, compares them with expected values, and detects outliers that may indicate anomalies. These detections are performed using machine learning methods within a dynamic environment.

3.2 Performance Comparison of Deep Learning Approaches for EVSE Anomaly Detection

In the following, it is important to summarize which machine learning and deep learning algorithms have shown promising performance in the field of anomaly detection for EVSE. Since the data involves time series and hardware-related measurements, the paper [Ali20], titled *Deep Learning-based Intrusion Detection System for Electric Vehicle Charging Station*, provides a valuable overview of suitable deep learning models in this context.

The authors compare a Deep Neural Network (DNN) with a Long Short-Term Memory (LSTM) model and conclude that the LSTM architecture achieves high accuracy—over 99%—after only a few training epochs. This supports the later decision in this thesis to select the LSTM model for better performance and anomaly detection in EVSE environments.

4 Data Acquisition

4.1 Data Sources for EVSE Anomaly Detection

In line with the approach described in the paper [EDBF24] *Enhancing EV Charging Station Security Using A Multi-dimensional Dataset : CICEVSE2024*, I recorded Hardware Counter Performance (HCP) kernel events and tracepoint events to classify anomalies in Electric Vehicle Supply Equipment (EVSE) data. For efficient monitoring, I also utilized the **perf** tool, a robust Linux performance analysis utility capable of capturing a variety of performance data, including hardware events, software events, and tracepoints [con25]. This tool offers flexible and effective monitoring of system performance and is instrumental in diagnosing issues efficiently.

To perform the data acquisition on a Raspberry Pi Zero W, I used the **perf.exporter**[Hod19] to record up to 900 Tracepoint-events, which were selected based on the CICEVSE2024 dataset. Initially, I developed a Python script to capture kernel and HCP events. However, when expanding to about 110 events, my developed script's runtime increased to approximately 9 seconds, which proved insufficient for real-time analysis. Consequently, I investigated the internal workings of the **perf** tool to understand how it manages specialized events and optimizes recording.

As detailed in *Linux perf event Features and Overhead*[Wea15], **perf** supports multiplexing, allowing the measurement of more events than there are available hardware counters—albeit at the cost of accuracy loss and increased system overhead. Recent updates can result in resource usage exceeding 20%, compared to approximately 2% in older tool versions. To minimize these inaccuracies, I grouped events per CPU, exploiting up to 8 available counter registers and limiting each group to a maximum of 6 events.

For the monitoring interval, I followed recommendations from the literature[Lim+14], which addresses multiplexing and accuracy trade-offs. The suggested minimum monitoring interval is around 10 ms for reliable results. Balancing the number of counter registers and striving for good measurement accuracy, I configured groups of 6 hardware events per group, with a monitoring interval of 0.1 seconds. This configuration yielded data with sufficient accuracy to detect anomalies in the EVSE dataset.

To further optimize efficiency and accuracy, I transitioned to developing the data acquisition script in Go. The Go programming language offers low-level hardware access and, with available libraries, integrates more seamlessly with the **perf** tool compared to Python. Additionally, the script exports all monitored hardware events from **perf** into a Prometheus database, enabling centralized and convenient access to all collected data. This setup streamlines subsequent analysis, visualization, and anomaly detection activities. The script is provided in the following section.

Here you can find two functions of the script the first one is used to format the data for Prometheus and the second one is used to monitor the data with the `perf` tool. The whole script you can find in the appendix 10.

```

1  // formatForPrometheus converts the filtered performance data into
   // Prometheus-compatible metrics format.
2  func formatForPrometheus(data string) string {
3      lines := strings.Split(data, "\n")
4      var result []string
5      for _, line := range lines {
6          parts := strings.Fields(line)
7          if len(parts) >= 2 {
8              // Clean up the metric name
9              metricName := strings.ReplaceAll(parts[1], ".", "_")
10             // metricName = strings.ReplaceAll(metricName, "-", "_")
11             // Remove decimal points from the number and parse it
12             metricValue := strings.ReplaceAll(parts[0], ".", "")
13             if value, err := strconv.ParseFloat(metricValue, 64); err == nil
14             {
15                 // Format value appropriately for Prometheus (integer or
16                 // float)
17                 if value == float64(int(value)) {
18                     result = append(result, fmt.Sprintf("%s %d", metricName,
19                     int(value)))
20                 } else {
21                     result = append(result, fmt.Sprintf("%s %f", metricName,
22                     value))
23                 }
24             }
25         }
26     }
27     return strings.Join(result, "\n")
28 }
29
30 // getPerfData runs 'perf stat' with the specified events for a short
   // duration (0.1s)
31 // and returns the raw output as a string.
32 func getPerfData(events []string) (string, error) {
33     eventList := strings.Join(events, ",")
34     cmd := exec.Command("perf", "stat", "-e", eventList, "--", "sleep", "
35     0.1")
36     output, err := cmd.CombinedOutput()
37     if err != nil {
38         return "", err
39     }
40     return string(output), nil
41 }

```

Listing 1: Go script for performance monitoring via perf for Prometheus

4.2 Integration of Prometheus

In the research on which database would be suitable for centralized data storage and analysis, **Prometheus** was a key focus. Not only is it an open-source project [20125], but it is also well known in the industry as a reliable monitoring solution for complex systems. It has a large community, along with comprehensive documentation and strong support, which are essential for maintaining a stable IT environment.

Another reason for choosing **Prometheus** is its seamless integration with Grafana for visualization, as well as its powerful PromQL query language. Additionally, **Prometheus** can store metrics in a time series database, which is particularly important for our detection systems [BMM24].

Prometheus consists of several components that are required for its operation. The core of the system is the **Prometheus** server, which collects and stores metrics and provides access to them through the PromQL query language.

Client libraries are directly integrated into the application code and expose a client-server interface. The **Prometheus** server periodically fetches data from them — by default every 15 seconds, although this interval is configurable.

In addition to the core components, **Prometheus** also supports optional components:

- **Push Gateway:** For external or short-lived metrics
- **Alertmanager:** For alerting and notifications
- **Exporters:** For monitoring additional systems, databases, or APIs
- **Visualization tools:** Such as Grafana

[SM24]

As explained in the previous subsection, **Prometheus** collects data via metrics exporters (see code snippet 3). After the data is collected and stored centralized in the **Prometheus** time series database, we developed a Python script that allows exporting the data within a defined time range and with specified time steps.

This export functionality is essential for enabling further analysis, including exploratory data analysis, using external tools such as Python or statistical software. The following code snippet illustrates the script used to export **Prometheus** data in CSV format:

```
1
2  //TODO update the skript ohne cpu aufteilung
3
4  PROMETHEUS_URL = "http://192.168.1.25:9090"
5  STEP = "10" # seconds
```

```

6  CHUNK_HOURS = 2
7
8  start_time = datetime(2025, 7, 18, 13, 36, 17, 210000)    # 210000 = 210ms
9  end_time = datetime(2025, 7, 18, 16, 0, 7, 210000)
10
11 def main():
12     chunks = build_chunks(start_time, end_time, CHUNK_HOURS)
13     data = defaultdict(dict)
14     timestamps = set()
15     values_dict = {}
16     # Mapping: (Feature, cpu_label) -> passender Querystring
17     metric_query_map = {}
18
19     for feat in FEATURES:
20         prom_name = feat.replace("-", "_")
21         url = f"{PROMETHEUS_URL}/api/v1/query"
22         hit = False
23
24         # 1. prom_name (ohne CPU)
25         params = {'query': prom_name}
26         try:
27             resp = requests.get(url, params=params)
28             resp.raise_for_status()
29             results = resp.json()['data']['result']
30             if results:
31
32                 for series in results:
33                     cpu_label = series['metric'].get('cpu', 'no_cpu_label')
34                     value = float(series['value'][1])
35                     values_dict[(feat, cpu_label)] = value
36
37                     metric_query_map[(feat, cpu_label)] = prom_name
38             hit = True
39         except Exception:
40             pass

```

Listing 2: Python script for exporting data from Prometheus in CSV format

Here you can see a short summarization of the key advantages of Prometheus, which make it a suitable choice for our data acquisition and live monitoring needs:

Advantage	Description
Powerful PromQL query language	Enables complex and efficient queries on metric data.
Centralized data storage	Stores all metrics centrally as time series data.
Extensive documentation and community support	A large community and comprehensive documentation ensure quick support and continuous development.
Seamless integration with Grafana dashboards	Provides easy integration with Grafana for visual dashboards and monitoring.
Open source and free	Free and open-source, offering broad availability and high customizability.

Table 4: Key advantages of **Prometheus**.
[BMM24][SM24]

5 Exploratory Data Analysis

5.1 Exploratory Data Analysis in the Context of EVSE Anomaly Detection

So far, we have provided an overview of the relevance and challenges involved in securing EVSE environments. We discussed the fundamental knowledge required before developing an anomaly detection system, gained insight into related work in this area, and explored how it can inform our own approach.

We have also examined how data can be acquired from EVSE environments and how **Prometheus** can be used to collect and store this data.

All of these sections and subsections are essential for understanding the context of our work and for showing how it is possible to develop an anomaly detection system for EVSE environments based on the analysis of hardware events and the collected data.

In this section, we focus on the Exploratory Data Analysis (EDA) of the collected data. EDA is a crucial step in understanding the data, identifying patterns, and preparing for further analysis or modeling. We use Python and its data analysis libraries within a Jupyter Notebook environment to visualize and explore the collected data.

This chapter is of particular importance because it defines the quality of the results in the following chapters and significantly influences how well anomaly detection using machine learning models will perform and how quickly they will learn. Therefore, we need to answer the following questions:

- Can we observe meaningful distributions in the features over time and in relation to attacks?
- Can we identify and define the most important features?
- Is there any noise in the data, and how significant is it?
- How strong are the correlations between the different features?

Answering these questions through Exploratory Data Analysis (EDA) provides a foundation for effective feature selection and model performance [Goo].

To convert the data into more useful formats, I used **Pandas**, a fast, powerful, and easy-to-use open-source data analysis tool for Python. It provides expressive data structures such as **DataFrame** and **Series** for working with structured and time series data.

I primarily used Pandas to convert my data, which was originally stored in **.csv** files, into **.parquet** files. Additionally, I used it to manipulate the dataframes by dropping cells, adding information such as "Attack" or "Benign" tags to almost 7000 rows. This preprocessing step was essential to clean the data before applying machine learning algorithms.

Some other useful key features of Pandas include:

- Tools for reading and writing data between various formats
- Flexible reshaping and merging of data
- Powerful grouping, aggregation, and transformation operations

[The25]

To create diagrams that help me better understand the data, I used a popular plotting library in Python called **Matplotlib**. Specifically, I used the `matplotlib.pyplot` module, which is a state-based interface to the Matplotlib library and provides a MATLAB-like way of creating static, animated, and interactive visualizations in Python.

Matplotlib is well-suited for quickly prototyping visualizations thanks to its simple and readable syntax. This allowed me to create a wide range of plots, including:

- **Line Plots:** Display data points connected by straight lines, useful for visualizing trends over time [Gee22d].
- **Scatter Plots:** Present individual data points as dots in a two-dimensional space, commonly used to show correlations between variables [Gee22e].
- **Bar Charts:** Use rectangular bars with variable lengths to represent values, making it easy to compare quantities across different categories [Gee22a].
- **Histograms:** Divide continuous data into bins and show the frequency or count of data points within each bin [Gee22c].
- **Heatmaps:** Visualize the relationship between multiple variables in a matrix format using color gradients [Gee22b].

These visualizations enabled me to better understand the data structure, identify patterns, and prepare the dataset for further processing.

I also used the Seaborn library it is like the other 2 libraries also an open-source Python library. It is exspecially know for making statistical graphiscs butilt on top of the matplotlib lib. Seaborn proves high-level interface for plotting informatikve and statititcal plots. It is also closely integrates with Pandas data structure what allows easy visualitzion of whole datasets and enables - Plotting of complex relationships - aumtomtac aggregation and statistical estimation (Means, confidence intervals, etc) - Thematic styling and enhanced color palettes. Therefore it was for me a perfect additon to the pandas and Mathplot libraries. [Wt25]

In addition to Matplotlib and Pandas, I also used the open-source Python library **Seaborn**. Seaborn is especially known for creating statistical graphics and is built on top of the Matplotlib library. It provides a high-level interface for drawing informative and attractive statistical plots. Seaborn is closely integrated with Pandas data structures, which makes it easy to visualize entire datasets. This integration enabled me to create complex visualizations with minimal code. Key features of Seaborn include:

- Plotting of complex relationships between multiple variables
- Automatic aggregation and statistical estimation (e.g., means, confidence intervals)
- Thematic styling and enhanced color palettes for better aesthetics

For these reasons, Seaborn was a perfect addition to the Pandas and Matplotlib libraries in my data analysis workflow. [Wt25]

Before I start with my own Feature selection I took a closer look at the selected HPC and Kernel features from the paper: Enhancing EV Charging Station Security Using A Multi-dimensional Dataset : CICEVSE2024 [EDBF24].

Event	Description	Backdoor	Crypto	Dos	Recon
instructions	Number of instructions executed	214.19	943.54	316.65	130.69
cache-misses	Number of cache misses	39.92	397.06	112.83	64.70
exc_taken	Exception taken	86.82	61.42	169.62	44.55
cpu-migrations	CPU Migrations	842.43	-100.00	1425.70	110.05
dTLB-store-misses	Data TLB - Write Misses	39.56	1068.57	101.48	72.97
l1d_cache_wr	Level 1 data cache access - Write	143.87	121.56	321.24	113.14
L1-icache-loads	Level 1 instruction cache access - Read	163.10	538.63	283.92	107.18
l2d_cache_rd	Level 2 data cache - Read	51.47	422.95	151.50	70.75
mem_access_rd	Data memory access - read	118.19	387.56	253.17	91.42
mem_access_wr	Data memory access - write	142.94	119.96	322.17	114.86
kmem_kfree	Kernel memory freeing event	149.06	67.66	2087.07	187.25
net_dev_xmit	Network device transmission event	27.42	33.79	172.88	59.64
qdisc.dequeue	Dequeue event	31.47	42.66	261.16	88.42
raw_syscalls_sys_enter	System call entry (raw) event	107.66	79.70	2.35	44.13
irq_softirq_raise	Software interrupt - Raised	90.29	65.72	1573.65	109.45
sched_migrate_task	Task migration event in the scheduler	670.49	-100.00	1091.74	78.18
sched_switch	Task switch event in the scheduler	42.10	39.22	332.82	24.59
syscalls_sys_enter_close	System call entry for close syscall	421.15	273.72	15.58	231.63
syscalls_sys_enter_read	System call entry for read syscall	44.32	30.19	37.09	56.21
syscalls_sys_enter_write	System call entry for write syscall	119.30	35.73	57.34	76.92

Figure 8: Selected HPC and Kernel Features from [EDBF24]

If we took at the timeline of the selected features we can already see some interesting patterns and trends especially for the (icmp-, tcp-, synonymous-) flooding attacks.

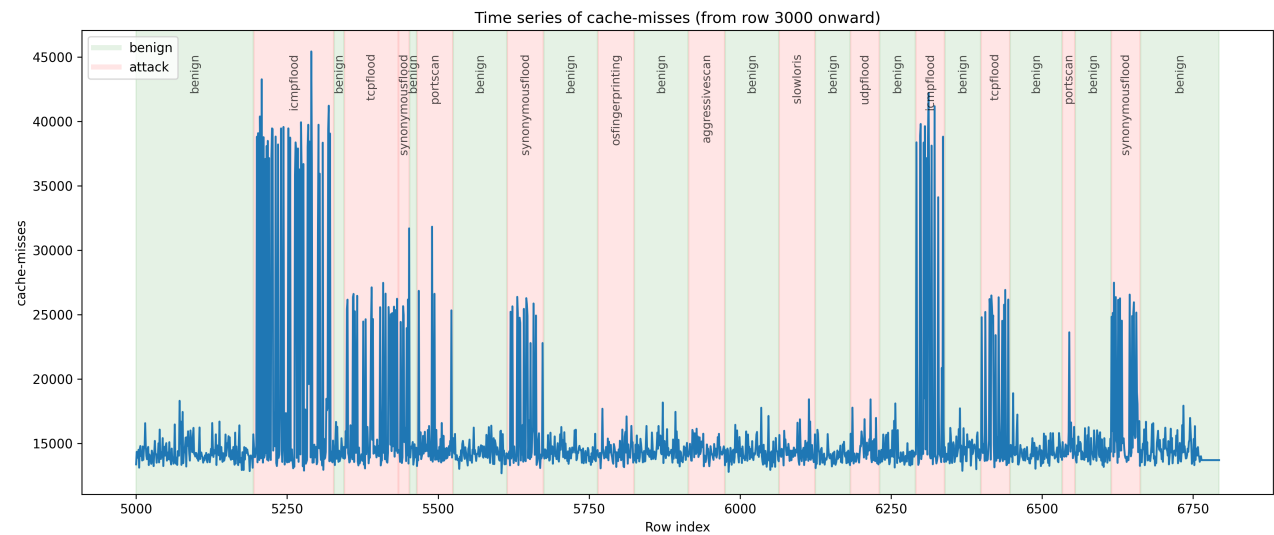


Figure 9: cache-misses timeline

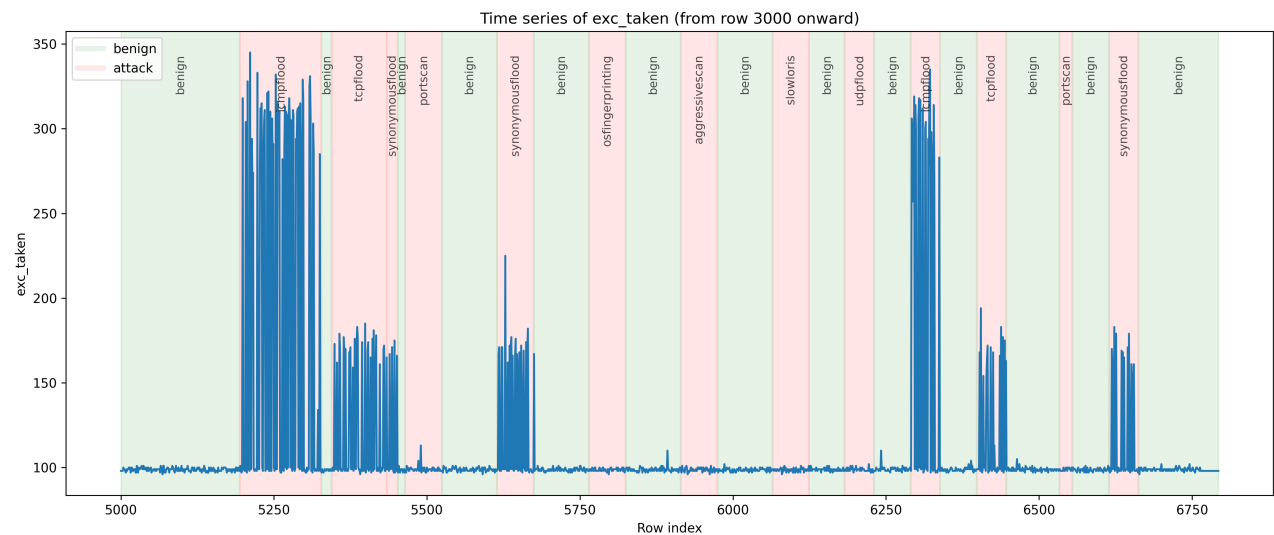


Figure 10: exc-taken timeline

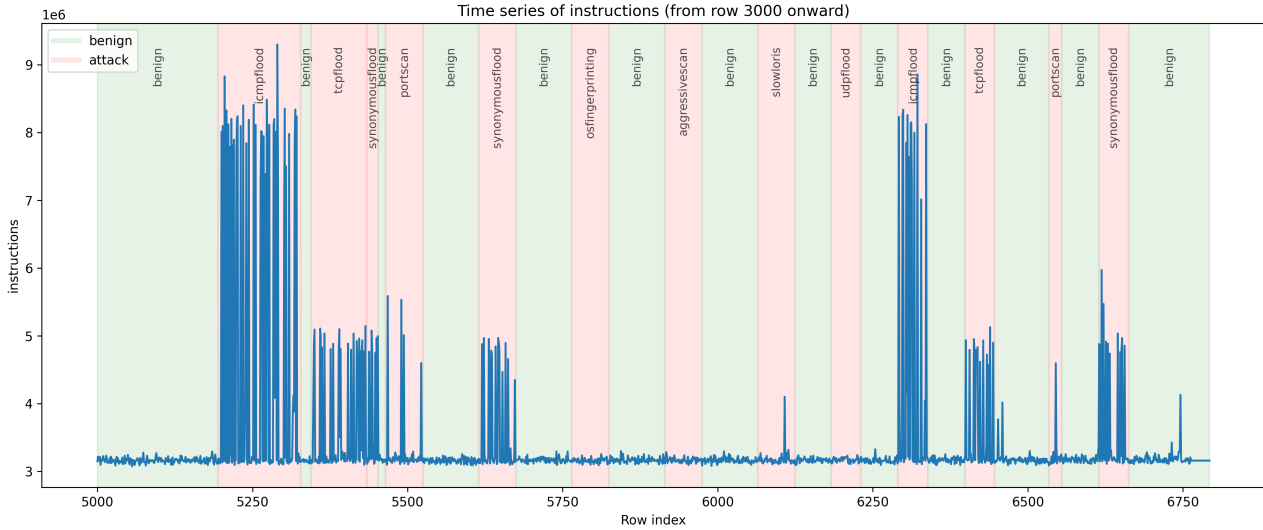


Figure 11: instructions timeline

Interestingly, the ICMP flood generates significantly more traffic across almost all measured features, such as *cache misses*, *exceptions taken*, and *instructions executed*. The ICMP flood attack is a type of Denial-of-Service (DoS) attack that overwhelms a target system with ICMP Echo Request packets, causing it to become unresponsive.

When compared to other attacks such as the TCP flood or the SYN flood, the ICMP flood almost doubles the observed values in several performance metrics. This can be explained by the fact that every single ICMP Echo Request must be fully processed and answered by the target system. As a result, this leads to increased memory accesses outside of the CPU caches, as well as more frequent interrupts within the kernel.

The paper "*An Experimental Detection of Distributed Denial of Service Attack in CDX 3 Platform Based on Snort*" also explains that the ICMP flood sends a significantly higher number of Echo Request packets, each of which requires an immediate response from the target system—unlike the TCP flood or SYN flood, which primarily rely on connection queues and do not necessarily require an immediate, full response.

5.2 Comparison of Public and Self-Generated Datasets

6 Feature Selection

6.1 Data Preprocessing steps

6.2 Key Feature Selection

6.3 Feature Selection Using Decision Trees

7 LSTM Architecture: Structure, Training, and Optimization

7.1 Structure and Functionality of the LSTM Architecture

7.2 Training Procedure and Validation Strategy

7.3 Parameter Configuration and Hyperparameter Optimization

8 Analysis and Evaluation

8.1 Experimental Setup and Evaluation Metrics

8.2 Results of Anomaly Detection Using the Machine Learning Model

8.3 Visualization and Interpretation of Detected Anomalies

9 Discussion

10 Conclusion/Future Work

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Anhang

```
1  package main
2  import (
3      "fmt"
4      "io/ioutil"
5      "net/http"
6      "os/exec"
7      "strconv"
8      "strings"
9      "time"
10
11     "gopkg.in/yaml.v2"
12 )
13
14 // Config represents the structure of the YAML configuration file.
15 // It includes high-level performance counters (HCP), software events (SW
16 // ), and hardware events (HW).
17 type Config struct {
18     HCP []string `yaml:"hcp"`
19     SW  []string `yaml:"sw"`
20     HW  []string `yaml:"hw"`
21 }
22
23 // Global variable to store the most recent performance data.
24 var perfData string
25
26 // getPerfData runs 'perf stat' with the specified events for a short
27 // duration (0.1s)
28 // and returns the raw output as a string.
29 func getPerfData(events []string) (string, error) {
30     eventList := strings.Join(events, ",")
31     cmd := exec.Command("perf", "stat", "-e", eventList, "--", "sleep", "
32         0.1")
33     output, err := cmd.CombinedOutput()
34     if err != nil {
35         return "", err
36     }
37     return string(output), nil
38 }
39
40 // parsePerfData filters and extracts only the relevant lines from perf
41 // output,
42 // based on the provided list of events.
43 func parsePerfData(data string, events []string) string {
44     lines := strings.Split(data, "\n")
45     var result []string
46     for _, line := range lines {
```

```
43     for _, event := range events {
44         if strings.Contains(line, event) {
45             result = append(result, line)
46             break
47         }
48     }
49 }
50 return strings.Join(result, "\n")
51 }
52
53 // formatForPrometheus converts the filtered performance data into
54 // Prometheus-compatible metrics format.
55 func formatForPrometheus(data string) string {
56     lines := strings.Split(data, "\n")
57     var result []string
58     for _, line := range lines {
59         parts := strings.Fields(line)
60         if len(parts) >= 2 {
61             // Clean up the metric name
62             metricName := strings.ReplaceAll(parts[1], ".", "_")
63             // metricName = strings.ReplaceAll(metricName, "-", "_")
64             // Remove decimal points from the number and parse it
65             metricValue := strings.ReplaceAll(parts[0], ".", "")
66             if value, err := strconv.ParseFloat(metricValue, 64); err == nil
67             {
68                 // Format value appropriately for Prometheus (integer or
69                 // float)
70                 if value == float64(int(value)) {
71                     result = append(result, fmt.Sprintf("%s %d", metricName,
72                     int(value)))
73                 } else {
74                     result = append(result, fmt.Sprintf("%s %f", metricName,
75                     value))
76                 }
77             }
78         }
79     }
80     return strings.Join(result, "\n")
81 }
82
83 // metricsHandler handles HTTP requests and returns the formatted
84 // performance metrics.
85 func metricsHandler(w http.ResponseWriter, r *http.Request) {
86     fmt.Fprintf(w, formatForPrometheus(perfData))
87 }
88
89 // chunkEvents splits the list of events into smaller chunks of a given
90 // size.
```

```
84 // This helps avoid issues with perf's limit on the number of
    // simultaneous counters.
85 func chunkEvents(events []string, chunkSize int) [][]string {
86     var chunks [][]string
87     for i := 0; i < len(events); i += chunkSize {
88         end := i + chunkSize
89         if end > len(events) {
90             end = len(events)
91         }
92         chunks = append(chunks, events[i:end])
93     }
94     return chunks
95 }
96
97 // validateEvent checks if a given performance event is available on the
    // system by using 'perf list'.
98 func validateEvent(event string) bool {
99     cmd := exec.Command("perf", "list")
100    output, err := cmd.CombinedOutput()
101    if err != nil {
102        fmt.Println("Error listing perf events:", err)
103        return false
104    }
105    return strings.Contains(string(output), event)
106 }
107
108 func main() {
109     // Load the YAML configuration file
110     config := Config{}
111     data, err := ioutil.ReadFile("config.yml")
112     if err != nil {
113         fmt.Println("Error reading config file:", err)
114         return
115     }
116     err = yaml.Unmarshal(data, &config)
117     if err != nil {
118         fmt.Println("Error parsing config file:", err)
119         return
120     }
121
122     // Validate and collect all available events from the config
123     var validEvents []string
124     for _, event := range append(config.HCP, config.SW...) {
125         if validateEvent(event) {
126             validEvents = append(validEvents, event)
127         }
128     }
129     for _, event := range config.HW {
```

```

130     if validateEvent(event) {
131         validEvents = append(validEvents, event)
132     }
133 }
134
135 // Split the events into manageable chunks (TODO: dynamically adjust
136 // based on CPU performance)
137 eventChunks := chunkEvents(validEvents, 6)
138
139 // Start a background goroutine to periodically collect performance
140 // data
141 go func() {
142     for {
143         var allPerfData []string
144         for _, chunk := range eventChunks {
145             data, err := getPerfData(chunk)
146             if err != nil {
147                 fmt.Println("Error:", err)
148                 allPerfData = append(allPerfData, "Error fetching data")
149             } else {
150                 allPerfData = append(allPerfData, parsePerfData(data,
151                     chunk))
152             }
153         }
154         // Combine data from all chunks into one string for Prometheus
155         // export
156         perfData = strings.Join(allPerfData, "\n")
157         time.Sleep(100 * time.Millisecond)
158     }
159 }()
160
161 // Start the HTTP server that exposes the '/metrics' endpoint for
162 // Prometheus scraping
163 http.HandleFunc("/metrics", metricsHandler)
164 fmt.Println("Starting server on :9011")
165 if err := http.ListenAndServe(":9011", nil); err != nil {
166     fmt.Println("Error starting server:", err)
167 }
168 }

```

Listing 3: Go script for performance monitoring via perf for Prometheus

content des beigefügten Datenträgers:

- ...
- ...

A Domändenmodell

Ein toller Anhang, der nicht nur als „*Müllhalde*“ genutzt wird, sondern in dem Bilder und contente auch mit eigenen Worten erklärt werden und den man auch für sich alleine lesen kann. Es sollten auch Referenzen auf die zugehörige ausführliche Behandlung im Hauptteil inklusive Seitenangabe mit `\pageref` gegeben werden.

Screenshot

Unterkategorie, die nicht im contentsverzeichnis auftaucht.